

Markscheme

Mathematics: analysis and approaches

Practice question bank

36. (a)

Consider the equation $\log_2 x + \log_4 x = 6$.

Find the value of x .

$$\text{use the change of base rule: } \log_4 x = \frac{\log_2 x}{\log_2 4} = \frac{\log_2 x}{2} \quad (M1)$$

$$\log_2 x \left(1 + \frac{1}{2}\right) = 6 \quad (M1)$$

$$\log_2 x = 4 \quad \mathbf{A1}$$

$$x = 16 \quad \mathbf{A1}$$

[4 marks]

(b)

Hence find the value of $\log_x 1024$.

$$1024 = 2^{10} \text{ and } x = 16 = 2^4, \text{ so } \log_x 1024 = \frac{10}{4} \quad (M1)$$

$$= 2.5 \quad \mathbf{A1}$$

[2 marks]